

Scalability & Future Plan

Date	15 March 2026
Team ID	SB-EDU-2026-04
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Reliance on cloud-based Gemini API	Latency and dependency on external service availability	High
2	Single-instance FastAPI deployment	Limited concurrent-user capacity before slowdown	High
3	No persistent database for user data	Learning history and quiz results lost each session	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	~20-30 concurrent users on a single Mac M1	Docker containers with multiple workers behind a load balancer, auto-scaling on cloud
2	Data Storage	No persistent storage; session-based only	Add PostgreSQL/MongoDB for profiles, quiz history & learning paths
3	Performance	Local LaMini-Flan-T5 inference on Mac M1	GPU-backed cloud endpoints with response caching
4	Security	No auth; API keys in plain env files	JWT authentication, secrets vault, API rate limiting

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Native mobile app (Android/iOS)	Q3 2026	Wider accessibility for students on the go
Phase 3	Multi-language support for content	Q4 2026	Reach non-English-speaking learners
Phase 4	Educator dashboard with analytics	Q1 2027	Enables classroom-level adoption